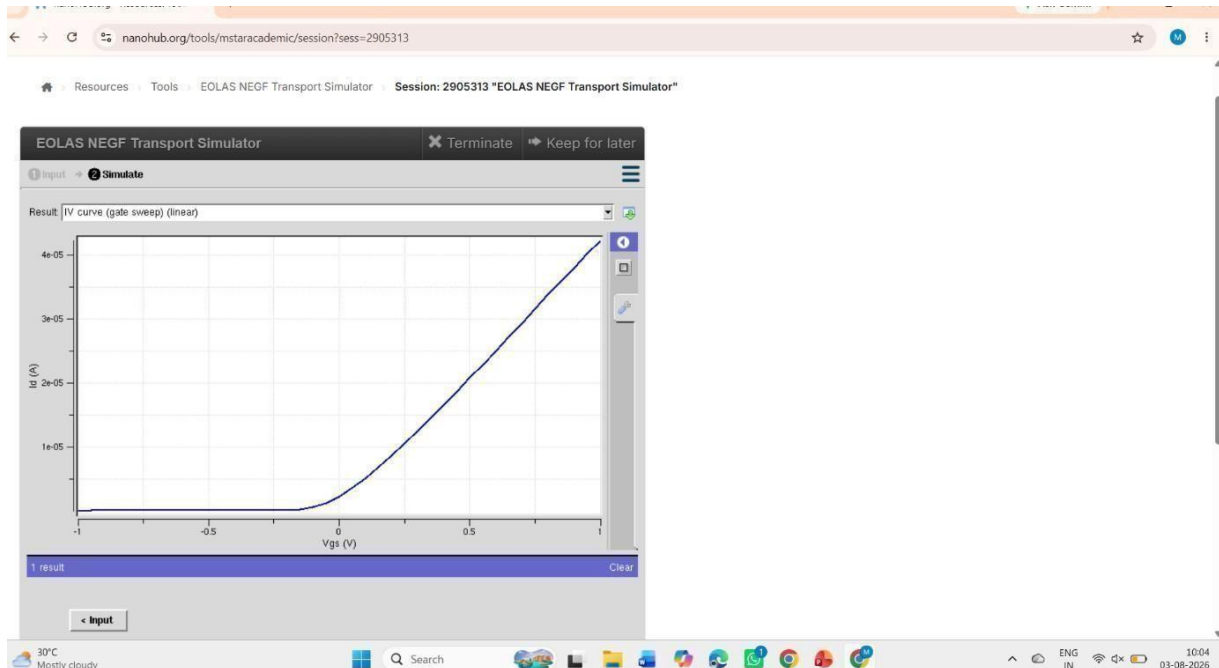


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Exp. 7. Simulation of VI Characteristics of FINFET, CNTFET and GFET using Nano Hub

Aim: To Obtain the VI Characteristics of FinFET by varying the Gate Channel length and oxide thickness

(a) FINFET



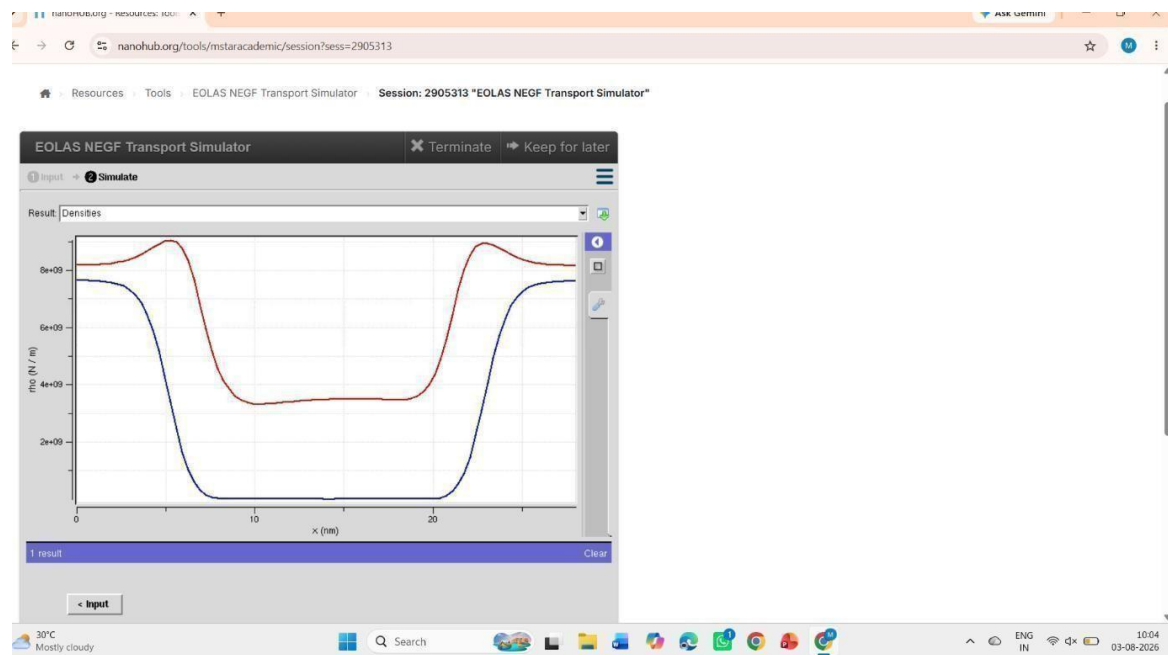
The graph represents the **transfer characteristics** obtained using the EOLAS NEGF Transport Simulator. The **drain current** is plotted against the **gate-to-source voltage**. At negative gate voltages, the current remains almost zero, showing that the device is in the **OFF state** with very few charge carriers available for conduction. As approaches **0 V**, the device begins to turn ON and the drain current increases gradually. For positive gate voltages, a much stronger increase in current is observed, indicating improved carrier transport through the device. At approximately **1 V**, the drain current reaches about **42 μ A**. The smooth increase in current demonstrates good **gate-voltage control and switching behavior**. Overall, the graph confirms that the simulated device exhibits typical **FET-like transfer characteristics**, with a low-current OFF region and a high-current ON region, making it suitable for transistor and electronic sensing applications.

Case Study 1: Simulate & obtain the current spectrum characteristics for the last bias point of FinFET . Analyze the performance for different Oxide layer thicknesses



The graph represents the **energy-resolved current characteristics** obtained using the EOLAS NEGF Transport Simulator. The **energy** is plotted on the Y-axis in electron volts (eV), while the **energy-resolved current** is plotted on the X-axis in A/eV. The graph shows that the current is mainly concentrated around the **Fermi energy region (eV)**, indicating that electron transport is strongly influenced by the available energy states near the Fermi level. At lower current values, the energy distribution extends to higher positive energies, while increasing current causes the energy range to move closer to the Fermi level. The gradual decrease in energy with increasing current indicates the contribution of different energy states to charge transport. Overall, the graph provides information about **energy-dependent electron transmission and current flow**, which helps in understanding the quantum transport behavior and conduction mechanism of the simulated nanoscale device.

Case Study 2: Simulate & obtain the electron density per sheet characteristics of a FinFET. Analyse its performance with different Gate region lengths

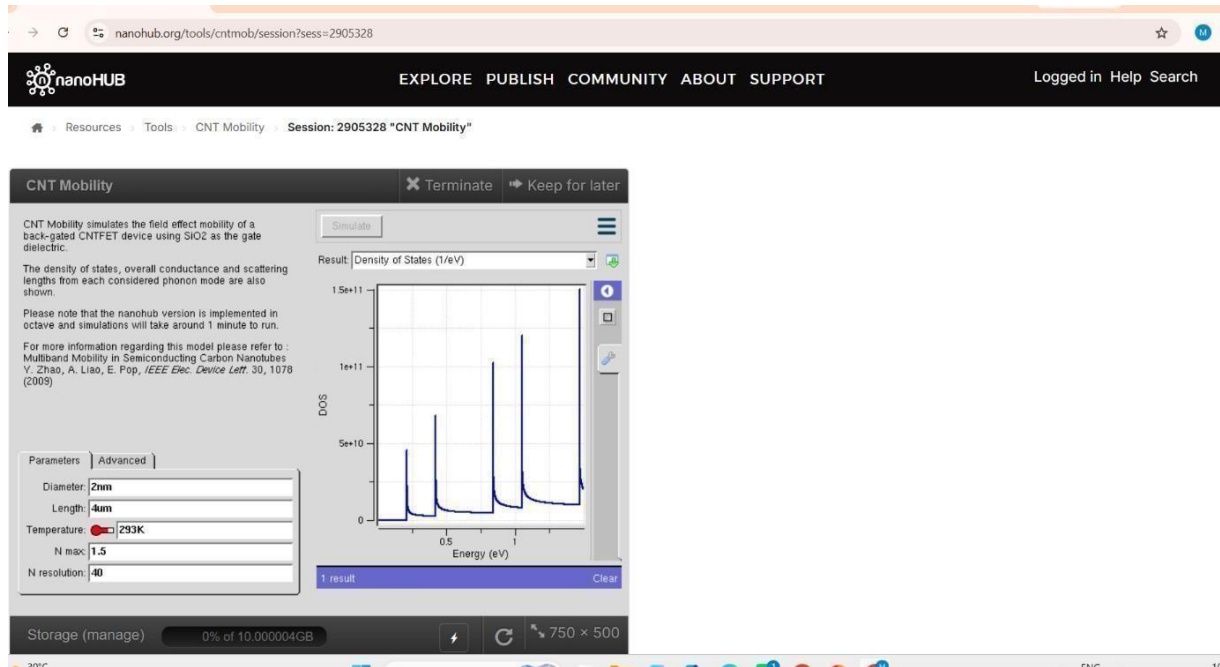


The graph represents the density distribution obtained using the EOLAS NEGF Transport Simulator. The electron density ($\rho(x)$) is plotted along the Y-axis in (N/m^3), while the position (x) along the device is shown on the X-axis in nanometers. The graph shows two density profiles that vary significantly across the device region. At the beginning and end of the device, the density is relatively high, while it decreases in the central region. The blue curve shows a stronger reduction in density, reaching nearly zero in the middle region, whereas the red curve maintains a higher density throughout the device. The changes near approximately 5–8 nm and 20–23 nm indicate transitions between different regions of the device. These variations in carrier density provide information about charge distribution, potential barriers, and electron transport within the nanoscale structure. Overall, the graph demonstrates that the electron density is not uniform and is strongly influenced by the device structure and transport conditions.

(b) CNTFET

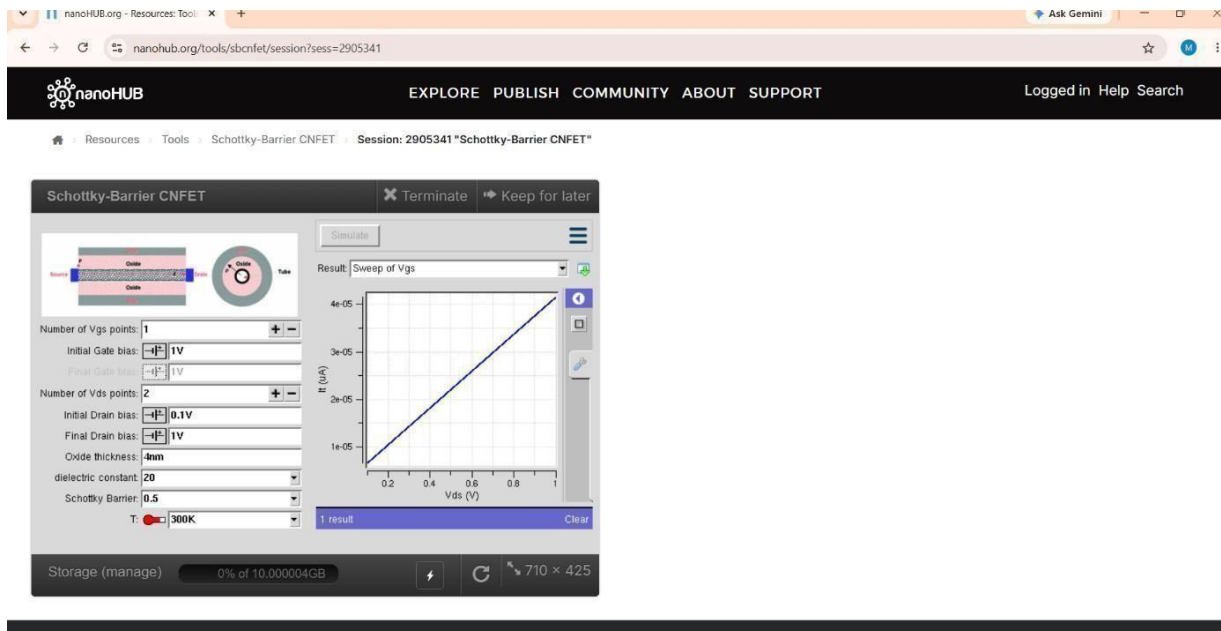
Aim: To obtain the density of states and conductance of CNTFET by varying the diameter and length.

Output:



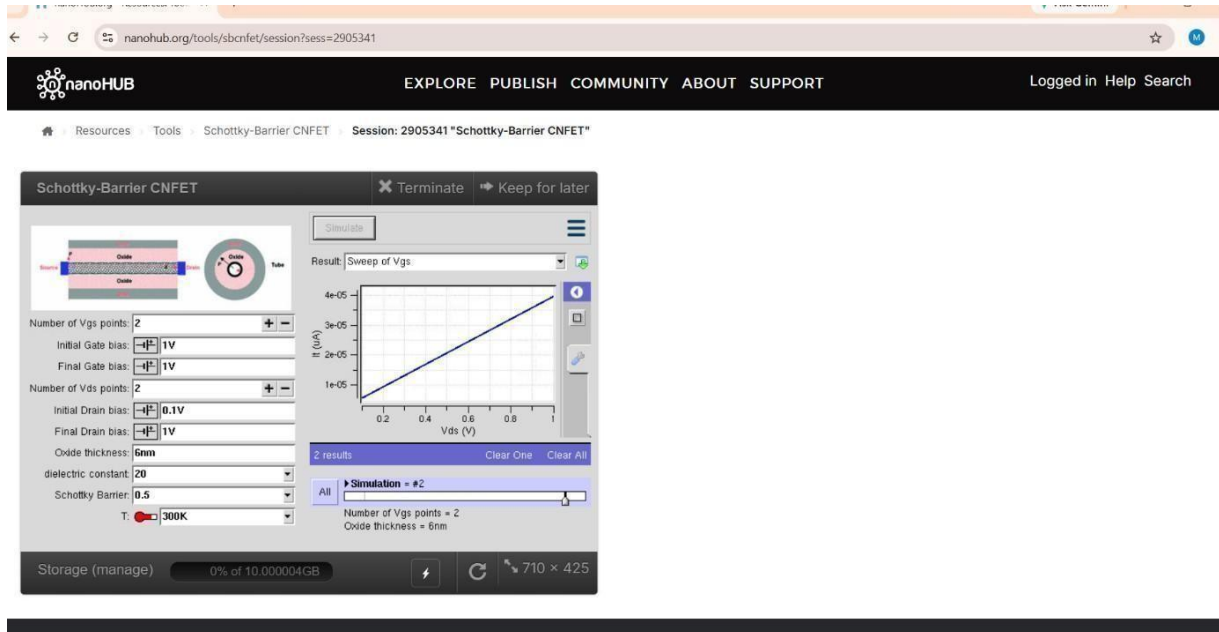
The graph represents the Density of States (DOS) versus Energy obtained from the CNT Mobility simulator. The energy (eV) is plotted on the X-axis, while the density of states (1/eV) is plotted on the Y-axis. The graph shows several sharp peaks at specific energy levels, indicating that a large number of available electronic states exist at these energies. Between the peaks, the DOS is relatively low, showing fewer available states for electron occupation. The increasing peak positions and their varying heights indicate the presence of discrete energy subbands, which are characteristic of the quantum-confined electronic structure of carbon nanotubes. These DOS peaks are important because they influence electron transport, conductivity, and carrier mobility in CNT-based devices. Overall, the graph confirms the quantized electronic behavior of the CNT and shows that its electrical properties strongly depend on the available energy states.

Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness



The graph represents the (I_D - V_{DS}) output characteristics of a Schottky-Barrier Carbon Nanotube FET (SB-CNFET) obtained using the nanoHUB simulator. The drain current (I_D) is plotted on the Y-axis in amperes, while the drain-to-source voltage (V_{DS}) is plotted on the X-axis in volts. The graph shows that the drain current increases almost linearly as (V_{DS}) increases from approximately 0.1 V to 1 V. The current reaches about (4.2×10^{-5}) A (42 μ A) at ($V_{DS}=1$) V. This linear behavior indicates that the device provides good current conduction and carrier transport under the applied bias. The characteristics are influenced by the Schottky barrier, oxide thickness, dielectric constant, and gate bias, which control the injection of carriers from the contacts into the carbon nanotube. Overall, the graph demonstrates the voltage-controlled transport behavior of the SB-CNFET and confirms its potential for nanoscale electronic and sensing applications.

Case study 2: Simulate & Obtain the VI characteristics of CNT-FET by varying CNT diameter

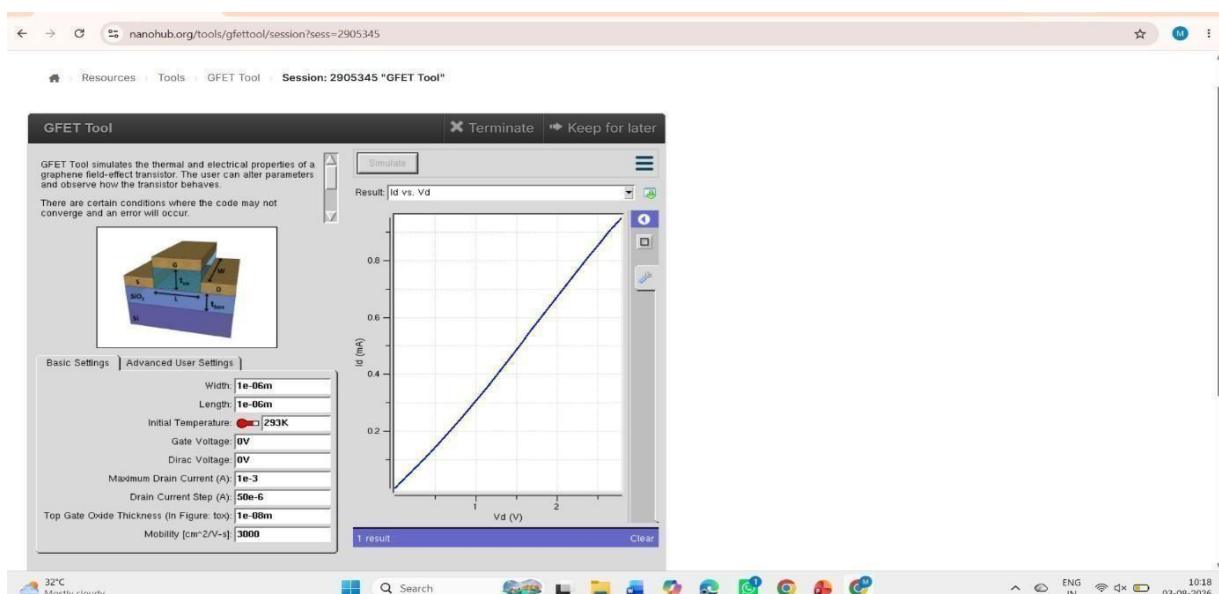


The graph represents the (I_D - V_{DS}) characteristics of a Schottky-Barrier Carbon Nanotube FET (SB-CNFET) obtained from the nanoHUB simulator with two gate-voltage points. The drain current (I_D) is plotted against the drain-to-source voltage (V_{DS}). The graph shows that the drain current increases steadily as (V_{DS}) increases from approximately 0.1 V to 1 V, reaching nearly (4×10^{-5}) A (40 μA) at 1 V. The nearly linear nature of the curve indicates good carrier transport and conduction through the CNT channel. The simulation uses an oxide thickness of 6 nm, dielectric constant of 20, Schottky barrier of 0.5 eV, and temperature of 300 K. Overall, the result demonstrates the voltage-dependent transport behavior of the SB-CNFET and shows that the device can provide significant drain current under increasing drain bias.

(c) GFET

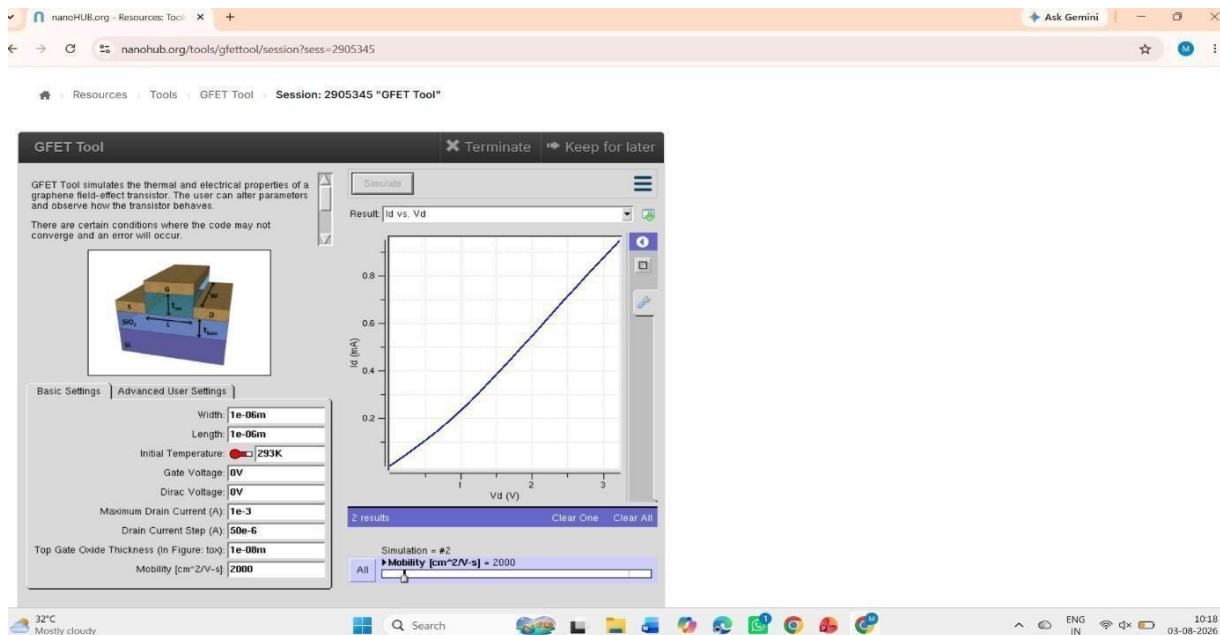
Aim: To simulate the Voltage-Current (VI) characteristics of a Graphene Field Effect Transistor (GFET) using appropriate simulation tools, analyze its performance under varying gate voltages and drain-source voltages, and understand its behavior as a high- mobility semiconductor device.

Case Study 1: How does varying the gate voltage (V_{gs}) affect the drain current (I_{ds}) in a GFET?



The graph represents the (I_D - V_D) characteristics of a Graphene Field-Effect Transistor (GFET) obtained using the nanoHUB GFET Tool. The drain current (I_D) is plotted on the Y-axis in mA, while the drain voltage (V_D) is plotted on the X-axis in volts. The graph shows an almost linear increase in drain current with increasing drain voltage, indicating good electrical conduction through the graphene channel. At a drain voltage of approximately 2.5 V, the drain current reaches about 0.9 mA. The linear behavior indicates that the device operates in a conductive region with effective carrier transport. The simulation uses a width and length of about 1 μ m, temperature of 293 K, zero gate voltage, and graphene mobility of approximately 3000 cm²/V·s. Overall, the result demonstrates the high electrical conductivity and efficient charge transport of graphene, making GFETs promising for high-speed electronic and sensing applications.

Case Study 2: What factors determine the maximum drain current ($I_{D,max}$) in a Graphene Field Effect Transistor (GFET)



The graph represents the (I_D - V_D) characteristics of a Graphene Field-Effect Transistor (GFET) obtained using the nanoHUB GFET Tool. The **drain current (I_D)** is plotted on the Y-axis in mA, while the **drain voltage (V_D)** is plotted on the X-axis in volts. The graph shows that the drain current increases continuously with increasing drain voltage. At lower (V_D), the current increases gradually, while at higher voltage the current rises more rapidly, indicating **nonlinear charge transport** in the graphene channel. The drain current reaches approximately **0.9–1.0 mA at around 3 V**. The simulation is performed at **293 K**, with **zero gate voltage**, and a graphene mobility of approximately **2000 cm²/V·s**. Compared with the previous result, the reduced mobility affects the current conduction and produces a different (I_D - V_D) response. Overall, the graph demonstrates the **voltage-dependent electrical behavior and effective carrier transport of the GFET**, confirming the suitability of graphene for nanoscale electronic and sensing applications.